EE 312
Experiment 7

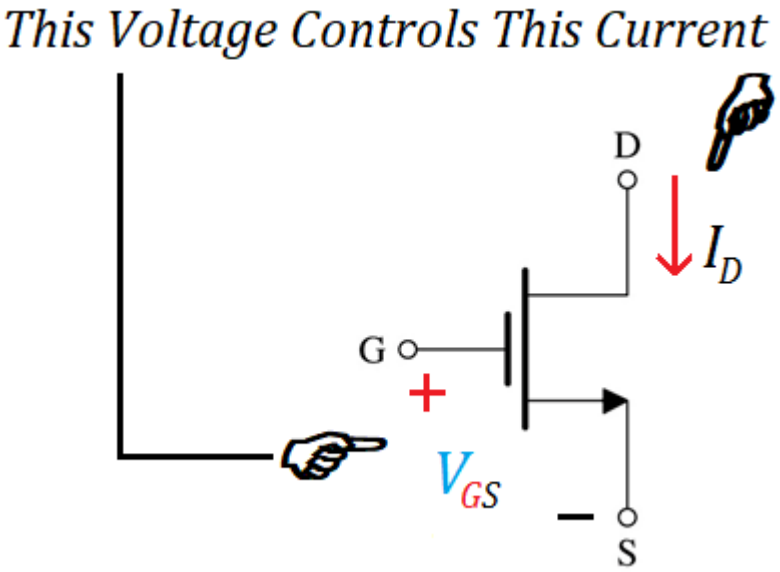
N CNANNEL ENHANCEMENT **MOSFET**

TURKI ALMADHI, LECTURER, DEPT. OF ELECT ENGINEERING.

What Is a Transistor?

It is a three-terminal device which *allows the voltage between two of its terminals to control the current flowing through the third terminal*.

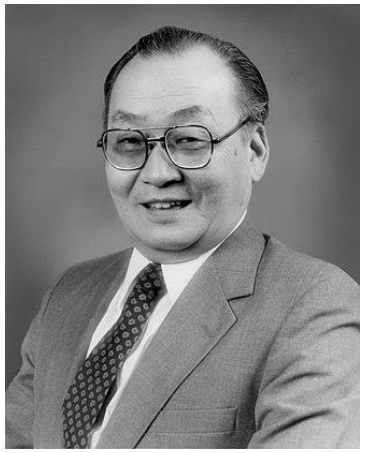




History

- MOSFET Stands for Metal Oxide Semiconductor Field Effect Transistor.
- It was invented by Mohamed Atalla (mech. engineer and self-taught physical chemist) and Dawon Kahng (elect. engineer) at Bell Labs in 1959.
- It is *the* most common type amongst semiconductor devices used today.



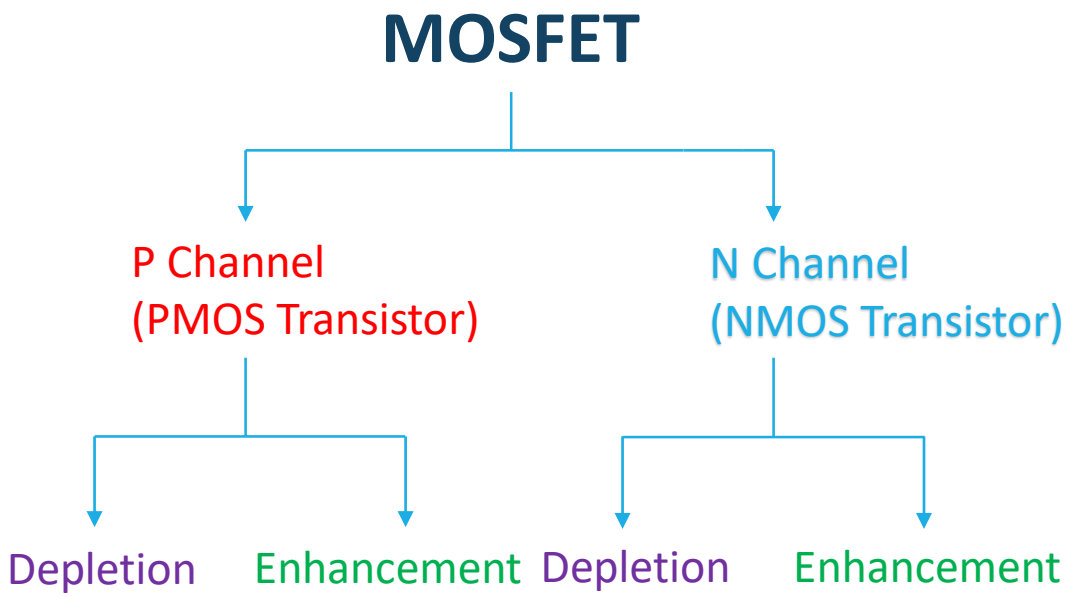


Behaviors Needed Often in Electronic Circuits

- We often need a device that can be *switched* on and off based on voltage.
- Also, we often need a device that can *convert* slight variations in the input voltage to directly proportional variations in the output current.

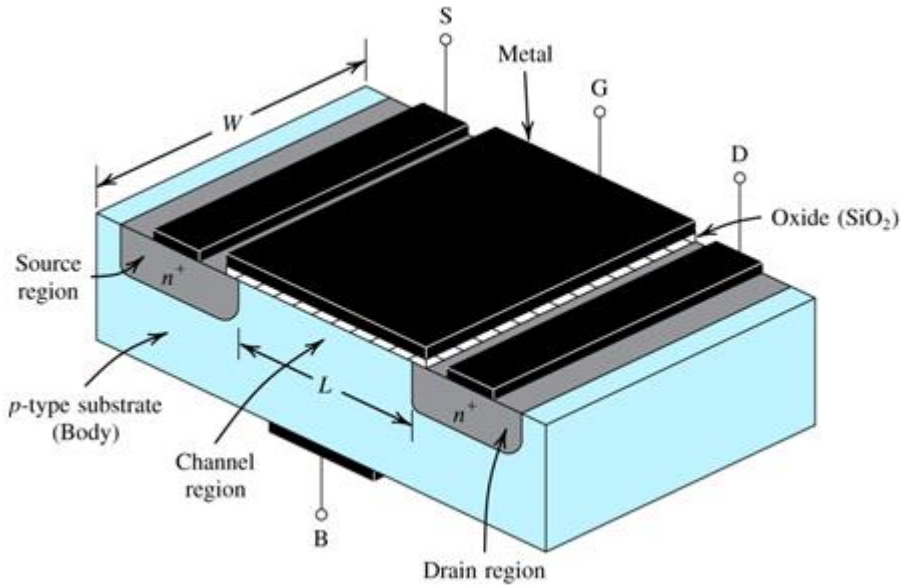
A Transistor Can Do the Above Functions Elegantly and Efficiently!

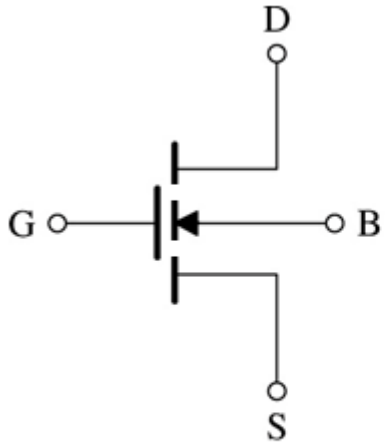
Types of MOSFETs



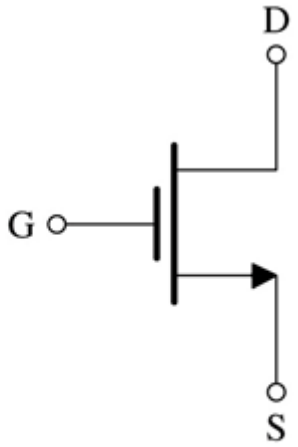
Structure and Symbol

- It has four terminals: Gate, Drain, Source and Body.





Used in ICs



Used in Discrete Transistots (Body Connected To Source)

Enhancement and Depletion

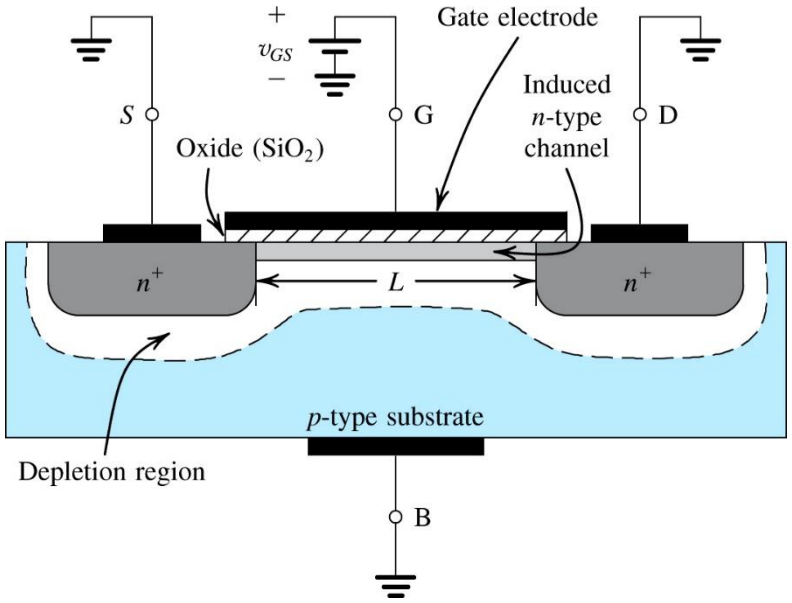
- Enhancement type MOSFET does *NOT* have a conducting channel to start with. A channel can be induced by applying a voltage of appropriate magnitude and polarity between the gate and source.
- Depletion type MOSFET DOES have a conducting channel to start with. A channel can be enhanced or depleted by controlling the voltage between the gate and source.

What is Meant By Field Effect?

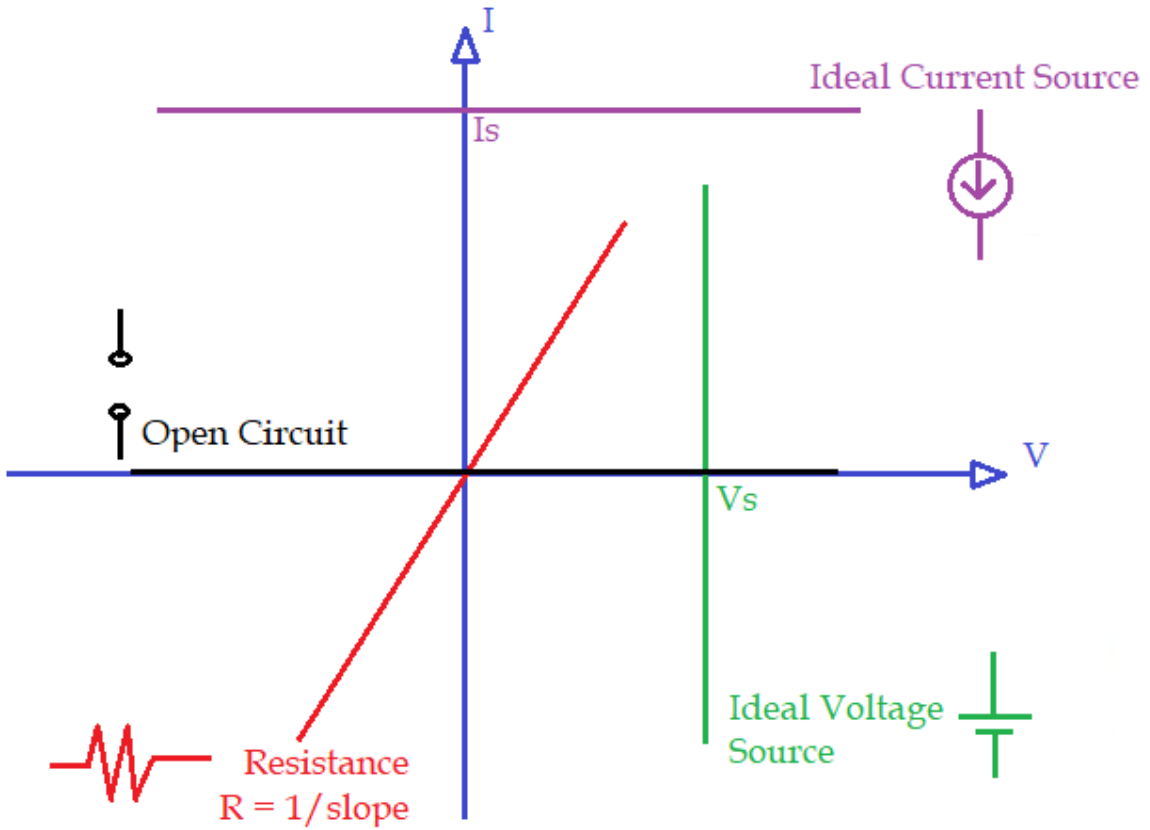
- For an Enhancement type MOSFET there is no conducting path between the drain and the source terminals of the MOSFET.
- A perpendicular electric field is used to invert the type of carriers in the substrate beneath the oxide such that they become compatible with the majority carriers in the drain and source to make a conducting path.
- That electric field is obtained by applying V_{GS} of appropriate magnitude and polarity.

Turning an E-MOSFET ON: The Threshold Voltage *Vt*

- When a large enough positive voltage (greater in magnitude than *Vt*) is applied to the gate of an E-NMOS transistor, an *n* channel is induced at the top of the substrate beneath the gate; otherwise the device is said to be cutoff ($I_D = 0$).
- A gate-to-source voltage large enough to make the electrons concentration in the channel equals to the holes concentration in the substrate is denoted *Vt,* the threshold voltage.

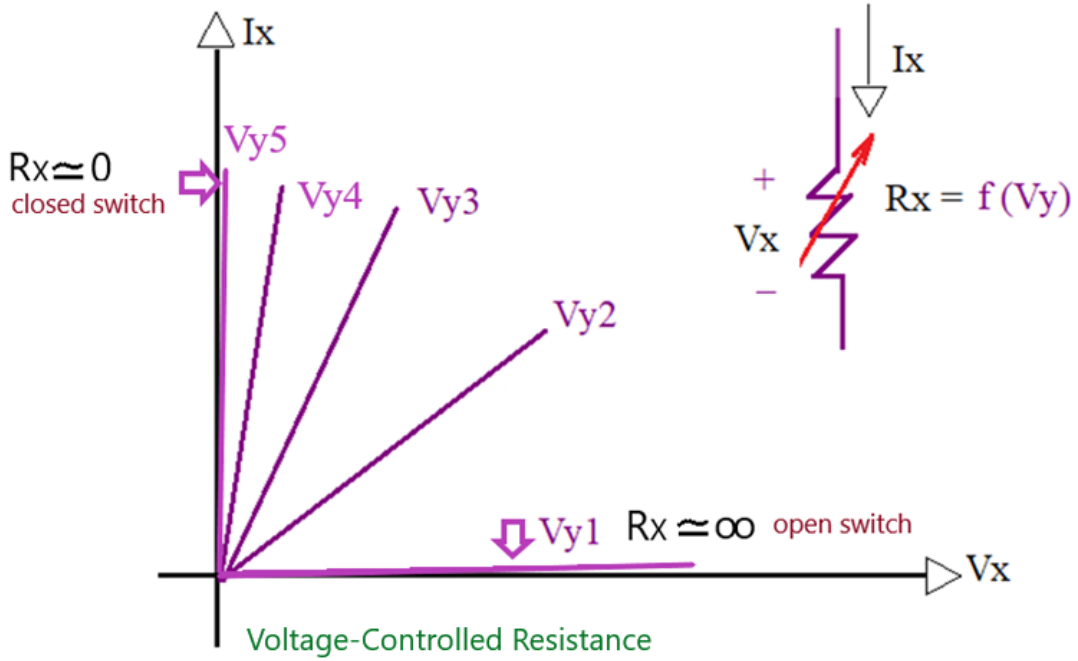


I-V Characteristic Curves for Common Elements Encountered in Linear Circuits



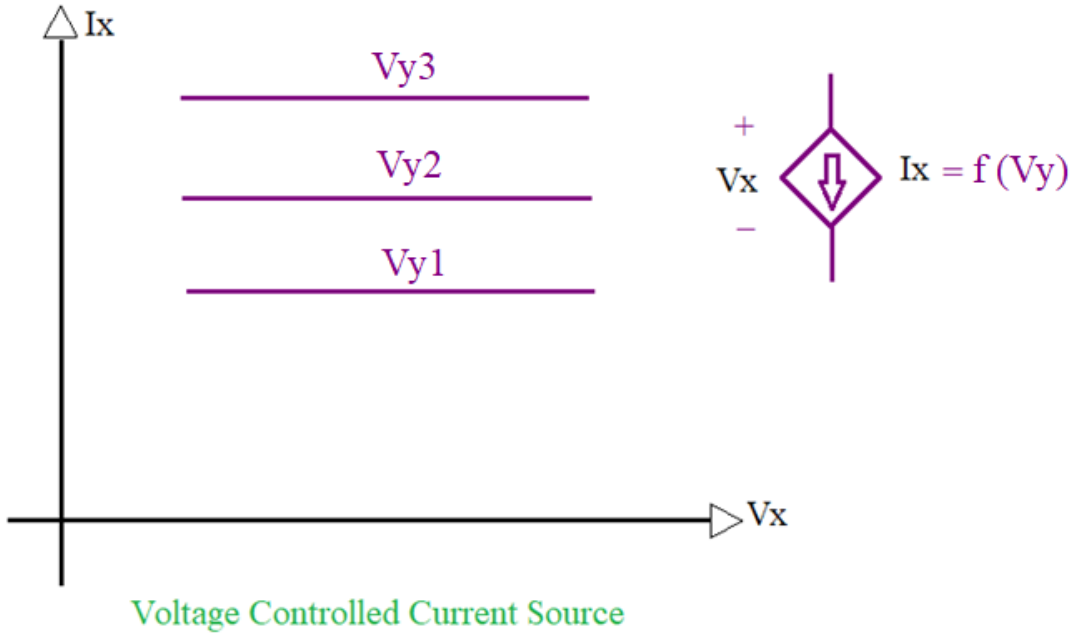
Voltage-Controlled Current Source and Resistance

I-V Characteristic Curves for a Voltage Controlled Resistance (Crucial for a *Controlled Switch*)



Voltage-Controlled Resistance

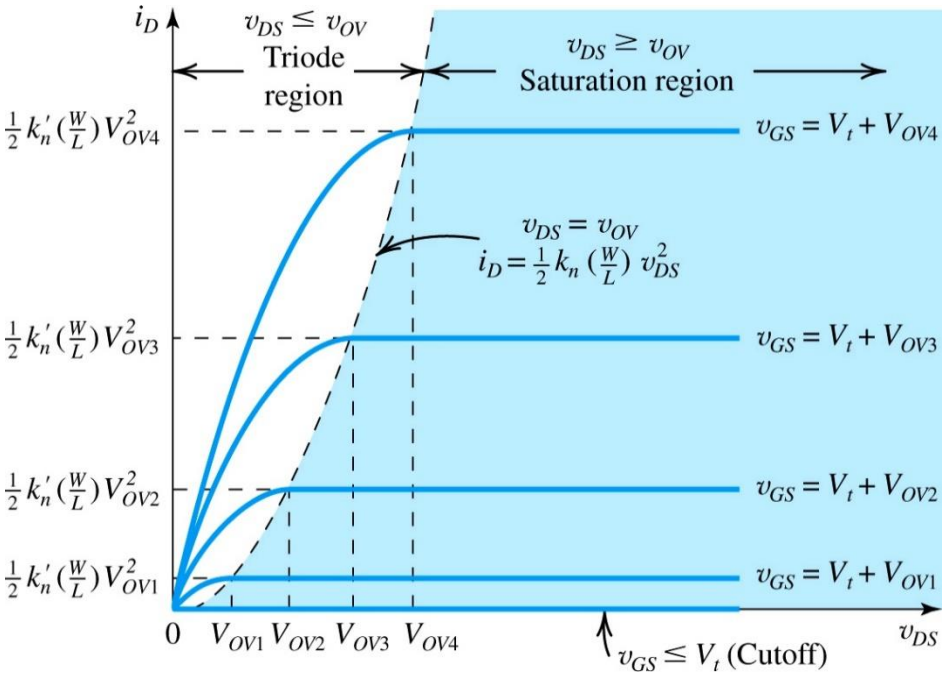
I-V Characteristic Curves for a Voltage Controlled Current Source (Crucial for an *Amplifier*)



Voltage Controlled Current Source

The Triode and Saturation Modes

- If a small positive V_{DS} (less than V_{OV}) is applied to the drain of an E-NMOSFET, the device operates in the triode region, where it acts as a voltage-controlled resistance. Note how (1/slope) → r_{DS} changes with V_{OV}.
- If V_{DS} is greater than or equal to V_{OV}, the device operates in the saturation region, where it acts as a voltage-controlled current source, VCCS. Note how the constant current value changes with V_{OV}.



Summary of the Modes of Operation and Relevant Applications

- If $v_{GS} < V_t$, the device is said to be **cutoff**.
- If $v_{GS} > V_t$, the device could be either in **triode** or in **saturation**, depending on the value of v_{DS} compared to the gate-to-source overdrive voltage $v_{OV} \equiv v_{GS} - V_t$.
- In the triode and cutoff modes of operation, the MOS transistor is used mainly as an **electronic switch**, which is very important for logic gates.
- In the saturation mode, and for sufficiently small signals, the MOS transistor acts as a linear voltage-controlled current source, and consequently can be used as an **electronic amplifier**.

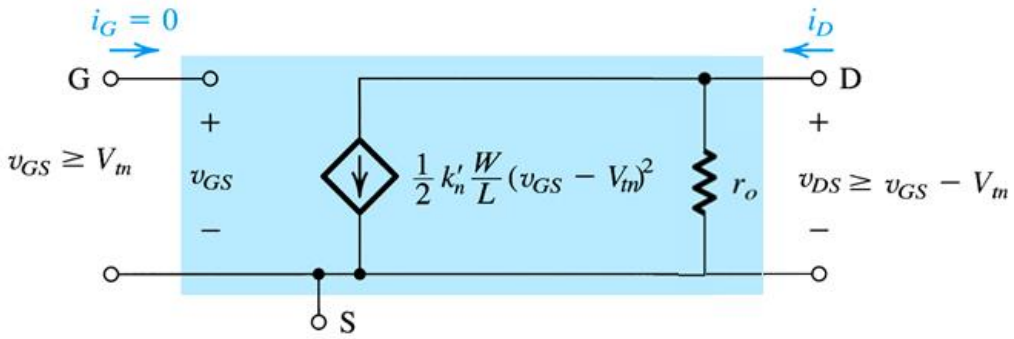
Which Voltage Affects ID The Most in the Saturation Mode?

Note that i_D is strongly controlled by v_{GS} and weakly by v_{DS} in the saturation mode, where it is given by:

$$i_D = \frac{k_n'}{2}\frac{W}{L}(v_{GS} - V_t)^2(1 + \lambda v_{DS})$$

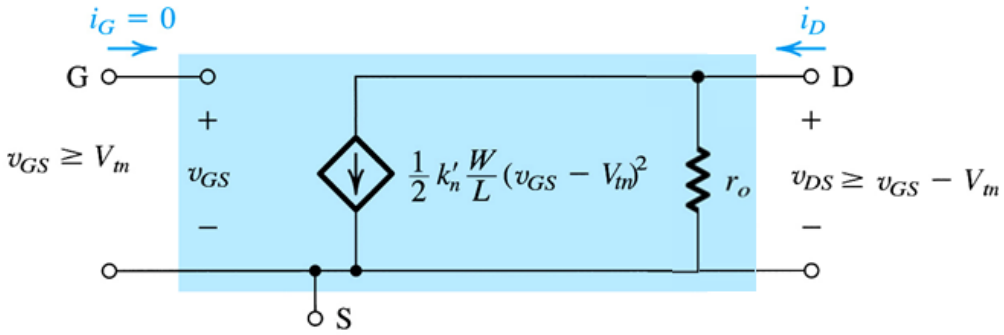
If $\lambda \approx 0 \rightarrow 1 + \lambda v_{DS} \approx 1$

$$i_D = \frac{k_n'}{2}\frac{W}{L}(v_{GS} - V_t)^2$$



Large Signal Equivalent Circuit Model

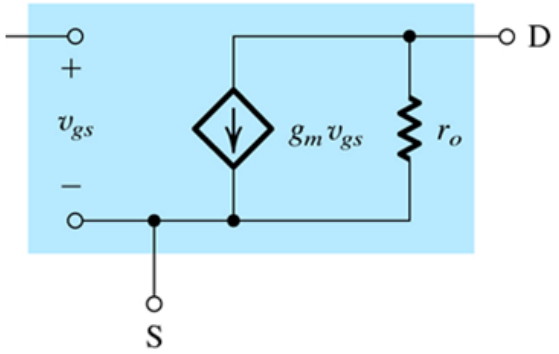
The Small Signal Approximate Model



Large Signal Equivalent Circuit Model

if v_{gs} is small enough

i.e., $v_{gs} \ll v_{GS} - V_{tn}$



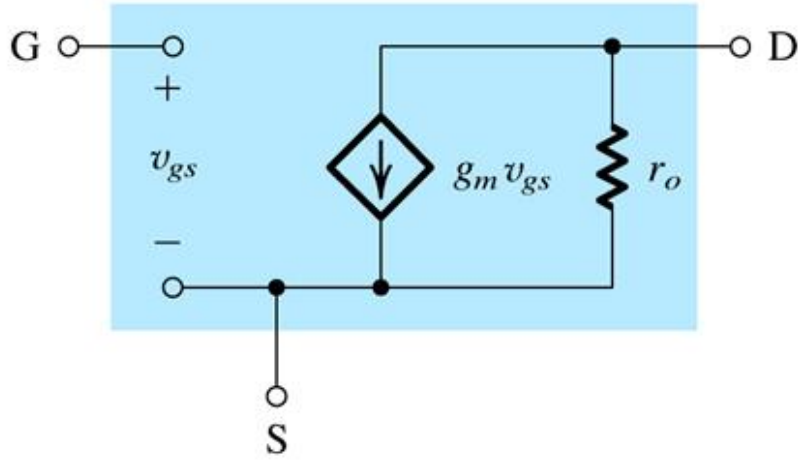
Small Signal Equivalent Circuit Model

Small Signal Parameters: g_m , r_o

- The transconductance g_m quantifies how much-in the saturation region- the drain current changes for a small change in v_{GS}.
- Caused by a phenomenon known as channel-length modulation, the weak dependence of i_D on v_{DS} in the saturation region at a constant V_{GS} is linear and can be modeled by *ro* between the drain and the source.

$$g_m \equiv \frac{\Delta i_D}{\Delta v_{GS}} = \frac{2I_D}{V_{GS} - V_t}$$

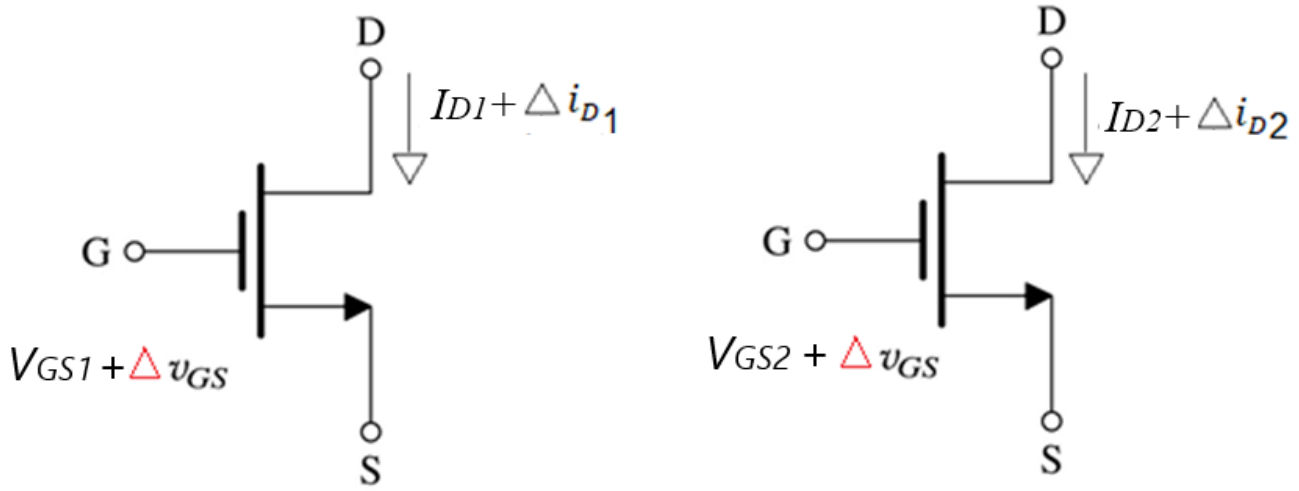
$$r_o \equiv \frac{\Delta v_{DS}}{\Delta i_D} = \frac{1}{\lambda I_D}$$



Which MOSFET Has a Greater Value of g_m?

Compare gm_1 to gm_2 if

$\triangle i_{D_1} > 2\,\triangle i_{D2}$



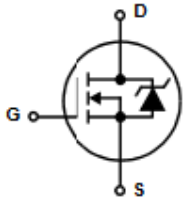
Switching Parameter: r_{DS}

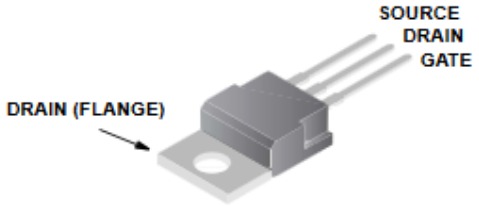
- In the triode region, the MOS transistor acts as a voltage-controlled resistance. Its closure resistance, denoted r_{DS} in textbooks and R_{ON} in datasheets, is inversely proportional to the slope of the i_D vs. V_{DS} relationship.
- For switching applications, e.g. logic gates, r_{DS} needs to be minimum for a closed switch → increase W/L or v_{OV}.
-

$$r_{DS} \equiv \frac{\Delta v_{DS}}{\Delta i_D} = \frac{1}{k_n' \frac{W}{L} V_{OV}}$$

MOSFET "IRF 620" Datasheet

IRF620
5.0A, 200V, 0.800 Ohm, N-Channel Power MOSFET



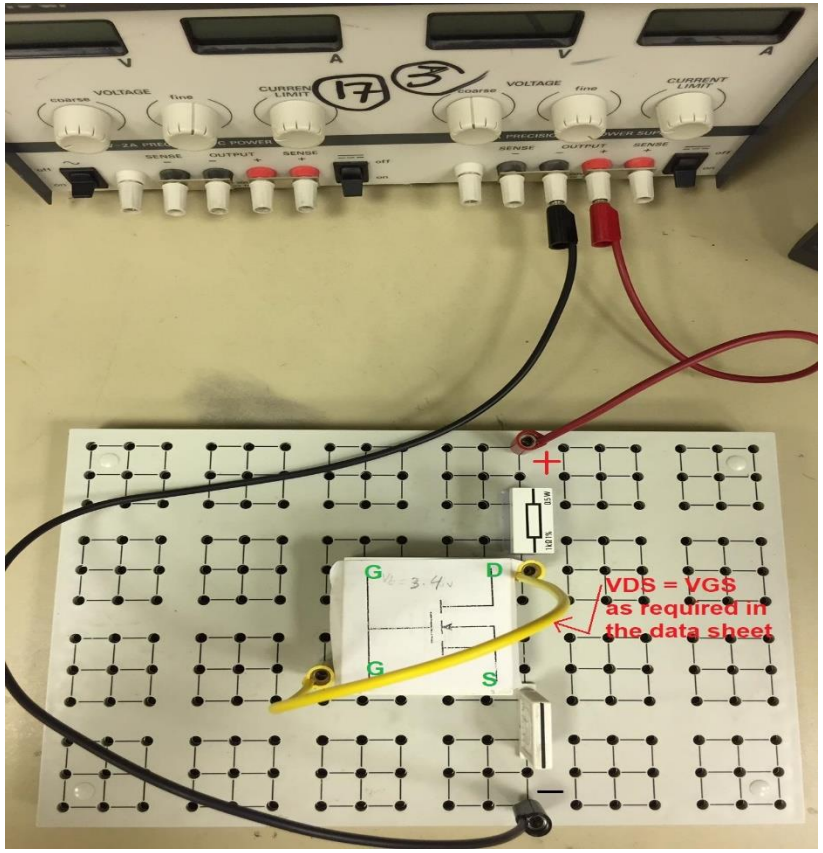


Electrical Specifications $T_C = 25^oC$, Unless Otherwise Specified

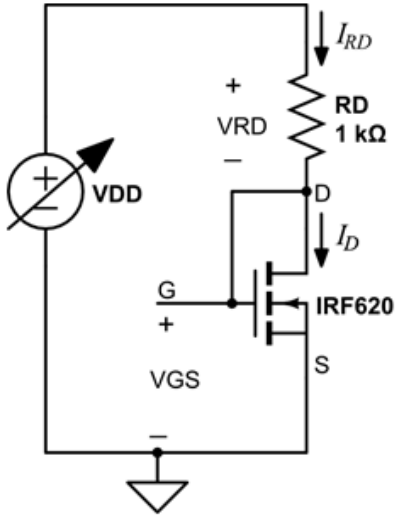
PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNITS
Drain to Source Breakdown Voltage	BV_{DSS}	V_{GS} = 0V, I_D = 250µA, (Figure 10)	200	-	-	V
Gate Threshold Voltage	$V_{GS(TH)}$	V_{DS} = V_{GS}, I_D = 250µA	2.0	-	4.0	V
Zero Gate Voltage Drain Current	I_{DSS}	V_{DS} = Rated BV_{DSS}, V_{GS} = 0V	-	-	25	µA
		V_{DS} = 0.8 x Rated BV_{DSS}, V_{GS} = 0V, T_J = 125°C	-	-	250	µA
On-State Drain Current (Note 2)	$I_{D(ON)}$	V_{DS} > $I_{D(ON)}$ x $r_{DS(ON)MAX}$, V_{GS} = 10V	5.0	-	-	A
Gate to Source Leakage Current	I_{GSS}	V_{GS} = ±20V	-	-	±100	nA
Drain to Source On Resistance (Note 2)	$r_{DS(ON)}$	V_{GS} = 10V, I_D = 2.5A, (Figures 8, 9)	-	0.8	1.2	Ω
Forward Transconductance (Note 2)	g_{fs}	V_{DS} > $I_{D(ON)}$ x $r_{DS(ON)MAX}$, I_D = 2.5A (Figure 12)	1.3	2.5	-	S

Measurement of The Threshold Voltage Vt

Procedure A



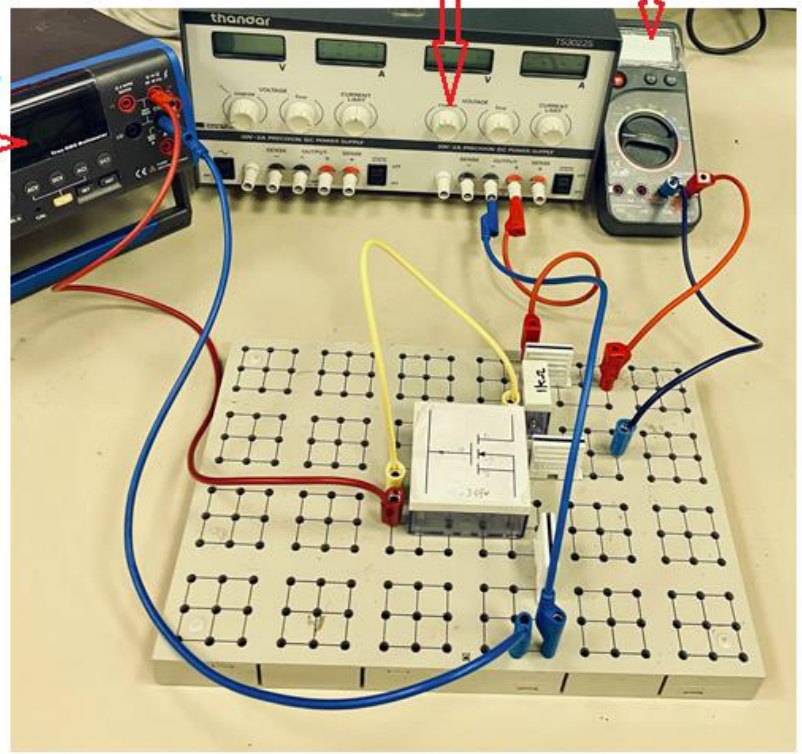
Increase V_{DD} until V_{RD} = 250 mV, then measure V_{GS}.

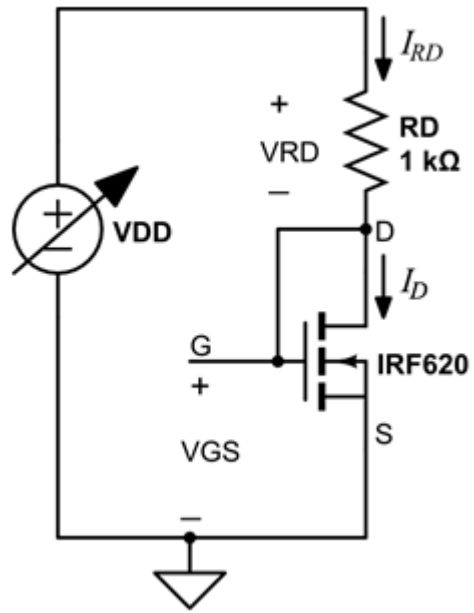


Setup with Multimeters Connected

1) increase V_{DD} until V_{RD} reads 0.25 v

2) read the corresponding $V_{GS} = V_t$





Measuring the i_D - v_{DS} Characteristics

Procedure B

